TechAxis

Learn First To Lead The Rest

A Microsyllabus on:

“Data Science with Python”

- Python with Data Science - Micro Checklist
- **Duration:** ~2 Months + 15 Days (~10 weeks)
- **Class Days:** Weekdays (SUN-FRI)
- **Daily Time:** 1 Hour

Class Start Date: _____

Trainer:_____

Week 1: Python Basics & Setup

- ☐ Day 1: Course Introduction & Python Installation (Anaconda, Jupyter, VS Code)
- ☐ Day 2: Python Basics - Variables, Data Types, I/O Operations
- ☐ Day 3: Operators (Arithmetic, Logical, Bitwise, Assignment)
- ☐ Day 4: Control Flow - If-Else, Loops (For, While)
- ☐ Day 5: Functions & Lambda Functions
- ☐ Day 6: Exception Handling & Debugging
- ☐ **Review on assignments & Discussions**

Week 2: Data Structures in Python

- ☐ Day 7: Lists - Operations, Slicing, Comprehensions
- ☐ Day 8: Tuples & Sets - Usage, Differences
- ☐ Day 9: Dictionaries - Key-Value Operations
- ☐ Day 10: String Manipulation & Regular Expressions
- ☐ Day 11: File Handling - Read, Write, Append
- ☐ Day 12: Python Modules & Importing
- ☐ **Review on assignments & Discussions**

Week 3: NumPy – Numerical Computing

- ☐ Day 13: Introduction to NumPy & Creating Arrays
- ☐ Day 14: Array Indexing, Slicing, & Reshaping
- ☐ Day 15: Broadcasting & Vectorized Operations
- ☐ Day 16: Mathematical & Statistical Operations in NumPy
- ☐ Day 17: Working with Missing Data in NumPy
- ☐ Day 18: Linear Algebra with NumPy
- ☐ **Review on assignments & Discussions**

Week 4: Pandas – Data Manipulation

- ☐ Day 19: Introduction to Pandas - Series & DataFrames
- ☐ Day 20: Importing Data (CSV, Excel, JSON)

- ☐ Day 21: Data Cleaning (Handling Missing & Duplicate Data)
- ☐ Day 22: Data Transformation (Sorting, Filtering, Applying Functions)
- ☐ Day 23: Grouping & Aggregation in Pandas
- ☐ Day 24: Merging, Joining, and Concatenating DataFrames
- ☐ **Review on assignments & Discussions**

Week 5: Data Visualization – Matplotlib & Seaborn

- ☐ Day 25: Introduction to Matplotlib - Basic Plots
- ☐ Day 26: Customizing Plots (Titles, Labels, Legends)
- ☐ Day 27: Subplots & Multiple Plots in Matplotlib
- ☐ Day 28: Introduction to Seaborn - Statistical Data Visualization
- ☐ Day 29: Pairplots, Boxplots, Violin Plots in Seaborn
- ☐ Day 30: Heatmaps & Correlation Matrices
- ☐ **Review on assignments & Discussions**

Week 6: Exploratory Data Analysis (EDA) & Preprocessing

- ☐ Day 31: Understanding Data - Types & Distributions
- ☐ Day 32: Handling Outliers - IQR
- ☐ Day 33: Feature Scaling - Normalization & Standardization
- ☐ Day 34: Encoding Categorical Data - One-Hot & Label Encoding
- ☐ Day 35: Feature Engineering & Feature Selection
- ☐ Day 36: Data Splitting - Train- Test Split
- ☐ **Review on assignments & Discussions**

Week 7: Web Scraping & Introduction to Machine Learning

- ☐ Day 37: Introduction to Web Scraping & Ethics of Scraping
- ☐ Day 38: Web Scraping with BeautifulSoup (Extracting HTML Data)
- ☐ Day 39: Navigating HTML Structure & Parsing Data
- ☐ Day 40 & 41: Saving Scraped Data into Pandas DataFrames
- ☐ Day 42: Mini Web Scraping Project – Extracting Data from a Website
- ☐ **Review on assignments & Discussions**

Week 8: Machine Learning (ML) – Supervised Learning

- ☐ Day 43: Introduction to Machine Learning - Basics & Applications
- ☐ Day 44: Supervised vs Unsupervised Learning
- ☐ Day 45: Linear Regression - Theory & Implementation
- ☐ Day 46: Logistic Regression - Theory & Implementation
- ☐ Day 47: Model Evaluation - MSE, RMSE, R^2
- ☐ Day 48: Overfitting & Regularization Techniques (Discussion)
- ☐ **Review on assignments & Discussions**

Week 9: Supervised & Unsupervised Learning

- ☐ Day 49 & 50 : Decision Trees - Concept & Implementation
- ☐ Day 51 & 52 : Random Forest - Ensemble Learning
- ☐ Day 53: Hyperparameter Tuning & GridSearchCV
- ☐ Day 54: Introduction to Cross-Validation
- ☐ **Review on assignments & Discussions**

Week 10: Unsupervised Learning & Advanced ML

- ☐ Day 55: Introduction to Clustering - K-Means Algorithm
- ☐ Day 56: Time-Series Analysis - Basics & Theory

Final Days: Deep Learning

- ☐ Day 57: Introduction to Deep Learning & Neural Networks
- ☐ Day 58: Basics of TensorFlow & Keras
- ☐ Day 59: Building a Simple Neural Network Model
- ☐ Day 60: Introduction to RNN (Neural Network)
- ☐ Day 61: RNN Application & Implementation
- ☐ Day 62: Introduction to LLM, GenAI & Prompt Engineering
- ☐ Day 63 & Day 64: **Projects internal discussions and completion**

Project & Career Guidance

- ☐ **Mini Project:** Web Scraping + Data Cleaning + Visualization
- ☐ **Final Project:** End-to-End Machine Learning Pipeline
- ☐ **Hands-on:** Model Deployment
- ☐ **Resume Building & Career Guidance**
- ☐ **Interview Preparation**
- ☐ **Final Presentation & Certification**

Coach/Trainer/Mentor Information:

Name: _____

Date: _____

TechAxis Authority:

Name: _____

Date: _____

Remarks: